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# Analyzing Geotag Patterns and Visitor Impact in Vålådalen Nature Reserve

## Assessing the potential of using crowdsourced geodata for monitoring visitor behaviour in protected Swedish landscapes

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Finally, to Finn, you light up my life little one.

**Abstract**

Multiple land use conflicts occur in Swedish Nature Reserves every year between the different land uses of; protection and restoration of valuable natural environments, meeting the need for outdoor recreation areas and Sámi reindeer herding communities. In recent years visitor numbers to these areas have risen, increasing the potential for conflict. Collecting relevant and accurate visitor data in protected areas is important to allow for efficient and pro-active management to take place. In 2024 the Swedish National Audit office (NAO) was critical about current monitoring and management standards within Swedish Nature Reserves. It is established in literature that crowd sourced geotagged photographs from social media can be used as a means of providing visitor counts, determining seasonal visitor patterns, and modelling relationships between visitors and infrastructure. The method examines if Flickr and Instagram geodata can determine visitor behaviour within Vålådalen Nature Reserve and if this aligns with current reserve management. It also examines if geotag activity intensity is associated with damage that can be identified from aerial photographs using an excess green index (ExG) analysis technique to compare nested buffers centered around the geographical mean of identified geotag clusters. The geotag distribution was found to align with the relative visitation rates in the zoning management plan from Länsstyrelsen Jämtland, The Flickr geotag dataset was also able to provide statistics on trail usage and seasonal visitor distribution rates. The analysis of geotag cluster points produced less conclusive results, with large variations between the sites and a weak relationship between increasing distance from a geotag cluster point and an increase in relative greenness. Overall the findings suggest that crowd-sourced geotag data can be analysed to identify visitor distribution and behaviour, however in this case the method was unable to identify a relationship between geotag clusters and ecological damage. The scope of this report was limited due to a lack of on-site data to compare with the patterns identified using geotags.

**Key words: Visitor behaviour, Visitor monitoring, Geotagging, Nature based tourism, Outdoor recreation**

# Contents

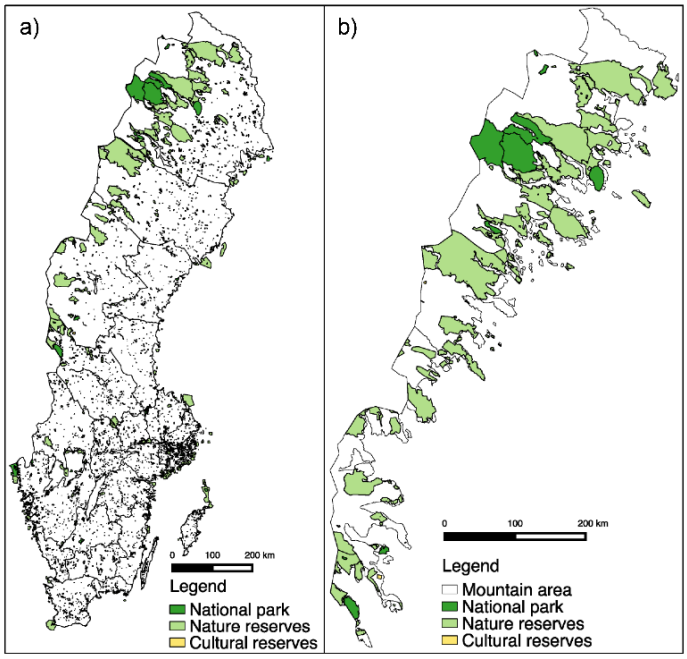
**1 Introduction ..... 6**
1.1 Background ..... 7
1.2 Orientation to the Research Area ..... 9
1.3 Research Question ..... 10
**2 Method ..... 11**
2.1 Study Area ..... 11
2.2 Data Collection and Initial Processing ..... 12
2.2.1 Flickr Geotag Data Collection ..... 14
2.2.2 Instagram Geotag Data Collection ..... 14
2.3 Distribution Analysis ..... 15
2.3.1 Zone Geotag Density Analysis ..... 15
2.3.2 Analysis of Geotag Proximity to Trails ..... 16
2.4 Human Impact Analysis ..... 17
**3 Results ..... 21**
3.1 Geotag Count and Density by Zone ..... 22
3.2 Trail Network and Geotag Proximity ..... 27
3.3 Human Impact Results ..... 28
**4 Discussion ..... 33**
4.1 Discussion of Results ..... 33
4.1.1 Geotag Distribution and Zoning ..... 33
4.1.2 Geotag proximity to Trail Buffers ..... 34
4.1.3 Human Impact ..... 34
4.2 Discussion of Method ..... 35
4.3 Limitations ..... 36
**5 Conclusion ..... 38**
**References ..... 39**
**Terms of publishing ..... 47**

# 1 Introduction

Across Sweden there are more than 5400 nature reserves (Figure 1), combined they form the greatest proportion of protected area in the country (Naturvårdsverket, 2026), covering 9.3% of Sweden's land area. Many of these protected areas are in the mountain regions and have sparse populations, however disputes occur frequently between different land use categories that exist within them, for example between traditional Sámi reindeer herding communities (RHC), tourists and tourism stakeholders and environmental protection agencies (Österlin et al., 2020).

**Figure 1**

*Mountain areas, National parks, Nature reserves and Cultural reserves in Sweden*




*Note:* a) Swedish National parks, Nature reserves and Cultural reserves. b) Mountain areas, National parks, Nature reserves and Cultural reserves in Northern Sweden. Adapted from "Different Worldviews as Impediments to Integrated Nature and Cultural Heritage Conservation Management: Experiences from Protected Areas in Northern Sweden" by Österlin, C., Schlyter, P., & Stjernquist, I. (2020), *Sustainability*, *12*(9), 3533. https://doi.org/10.3390/su12093533 Licensed under CC BY 4.0

In law, the Swedish Environmental Code SFS (1998:808) prescribes the reasons for the creation of these protected areas (Nature reserves), citing that

they should be used for the protection and restoration of valuable natural environments and meeting the need for outdoor recreation areas (Naturvårdsverket, 2026). The right of the Sámi to practise reindeer herding is protected under the Reindeer Husbandry Act (Ch. 1 & 2 Constitution Act 1974:152).

It is well established that managing the implementation of the different purposes and land uses, within individual nature reserves, requires careful monitoring and planning to protect the interests of all stakeholders and reduce conflict (Ankre, Fredman and Lindhagen, 2016).

## 1.1 Background

Swedish outdoor and mountain tourism has long been culturally significant, with thousands of people travelling to Swedish mountain areas annually. This is driven in part by the ideals of friluftliv (free-air-life) and allamansrätt, (The right of public access), both are considered core cultural values in Sweden (Saito et al. 2022). Increasing urbanization, health benefit promotion, and investments in outdoor (nature based) tourism opportunities in recent years have driven an increase in this interest in outdoor recreation (Ankre, Fredman and Lindhagen, 2016). The COVID-19 pandemic is also shown to have influenced an increase in the number of visitors to outdoor areas, and Hansen *et al* (2022) report that the number of people visiting nature at least once a week rose from 77 to 85% over the course of the pandemic.

The rise of smart phones and social media has also played a role in influencing visitor behaviour in outdoor spaces. The ease of capturing experiences and sharing them with wider audiences, including geotagging (adding geographical identification metadata to media), has increased significantly since the early 2000's, and social media sites such as Instagram, Facebook and Flickr have been shown to popularize certain destinations. (Mackenzie et al., 2023).

Increased visitor numbers to environmentally significant environments have been shown to impact these protected areas with key "icon" sites being particularly affected (Hadwen, Hill and Pickering, 2007). Damage to these sites can take place in the form of biological, chemical and physical alterations, with visible damage including fire scorched earth, trail degradation, exposed tree roots and ground down vegetation. Invisible

damage includes changes in nutrient pathways, fragmented ecosystems and proliferation of invasive species (Buckley, 2003). Alpine areas are particularly sensitive to damage, due to decreased ability for vegetation to recover from disturbance such as trampling or scorching (Leung et al., 2010).

As visitor numbers increase in Swedish Nature Reserves, so does the potential for conflict to take place. Sweden has a long history of nature protection, and was the first European country to form protected areas (initially national parks), from 1909 onwards, with the first nature reserves being introduced in the 1960's (Naturvårdsverket, 2026). In the management of these areas, Swedish strategies have historically placed focus on maintaining "naturalness" and preferred a hands-off approach to intervention. In more recent years, themes have emerged channelling increased intervention, particularly in the practice of restoring natural values (Steinwall, 2015).

In 2018 Naturvårdsverket updated the guidance document "Vägledning om förvaltning av skyddad natur" on protected area management. This document details practices for Nature Reserves, National Parks and Natura 2000 areas, covering roles, responsibilities, prioritization, safety and the management of visitors. While Naturvårdsverket has overreaching responsibility for the management of mountain areas, the country administrative boards (Länsstyrelsen) are responsible for the practical administration within their own municipalities (Naturvårdsverket, 2018).

Naturvårdsverket (2026), describes that an essential part of management of protected areas is an understanding of the values and interests that are important to visitors. Visitor behaviour is influenced by a range of factors including, accessibility, infrastructure and icon site location, it is never homogeneous throughout an area (Hadwen, Hill and Pickering, 2007).

To develop strategies to minimize ecological impact from tourists, and mitigate potential conflict, while maintaining positive visitor experiences, both visitor numbers and the behaviour of visitors and patterns of activity within protected areas must be known. Visitor data can be collected through manual methods, such as trail counters, observations, and surveys, alongside digital methods, such as tracking (Walden-Schreiner et al., 2018).

Due to factors such as funding and methodological inconsistencies, data collection is often limited and the lack of reliable data in remote protected areas can be a limitation for effective proactive management (Hadwen, Hill and Pickering, 2007).

Ankre, Fredman and Lindhagen (2016) report that Sweden lacks a nationally standardized system for visitor monitoring across nature reserves. In 2024 the Swedish National Audit Office published the report Förvaltningen av skyddad natur (RiR 2024:11), where the management of protected areas under the supervision of Naturvårdsverket was criticized. The report specifically notes that management is being impeded by ineffective monitoring systems, describing the current system as, "not fit for purpose", and highlighted that many of the current nature reserve and national park management plans are out of date (Riksrevisionen, 2024).

Without effective management such as appropriate infrastructure development, distribution of municipal funds and the pre-emptive protection of sensitive environments, the natural and recreational values in Swedish Nature Reserves risk being degraded and lost (Ankre, Fredman and Lindhagen. 2016), (Riksrevisionen, 2024).

## 1.2 Orientation to the Research Area

As social media has increased in popularity, the quantity of crowd-sourced data available for analysis has risen con-currently. A form of data that is accessible and relevant for the geo-spatial analysis of visitors in outdoor spaces, is volunteered geographical information, in the form of geotags. This geographical identification metadata is available alongside photographs on some social media websites and applications (Junker et al., 2017). Geotag metadata can be used to monitor visitors in several ways, and is increasingly being used to complement traditional visitor monitoring techniques in outdoors spaces, rising as an efficient technique that provides data from remote locations (Levin et al., 2017), (Tenkanen et al., 2017).

Social media outlets such as Instagram, Facebook and Flickr allow users the option of providing geodata when they upload photographs to their website or application. As global positioning system (GPS) technology has merged

with mobile telephones and become both publicly available and widely accessible, this practice has increased in frequency, with crowdsourced social media geodata becoming available for many locations worldwide. An estimated 11.5% of all geotagged photographs (20 million) available on Flickr are taken within protected outdoor spaces, (Levin et al., 2017, Walden-Schreiner et al., 2018).

Wilkins et al. (2020), reviewed 58 papers that analysed crowd sourced social media geodata in protected areas worldwide, and concluded that the temporal and spatial resolution of the data is often high enough to allow for it to be used to explore patterns of visitation in these areas.

Studies evaluating the effectiveness and accuracy of using crowd-sourced geotag data to monitor visitors in protected nature areas find that the data is useful both by itself as well as alongside traditional methods. The data can be used to provide both wider and site-specific information, model visitor flows, provide visitor counts, determine seasonal visitor patterns, and model relationships between visitors and infrastructure (Walden-Schreiner et al., 2018), (Wilkins et al., 2020).

## 1.3 Research Question

The purpose of this study is to explore the potential for crowd-sourced geotagged data to be used as a means of monitoring visitors in remote protected areas in Sweden. It will evaluate if the data can indicate visitor hotspots, movement patterns and areas of concentrated use in Swedish Nature Reserves. The aim is to contribute to the discussion surrounding the effective monitoring and sustainable management of protected areas in Sweden, with attention to how analysis of geotagged photographs can provide insight into geospatial visitor behaviour patterns and how these may influence land-use conflict mitigation between recreation, conservation and the Sámi RHC's in the area.

This research investigates if crowd-sourced geotagged data can be used to determine the behaviour of visitors in Swedish Nature Reserves, and if there is a relationship between the intensity of geotag activity and human impact within these protected areas.

# 2 Method

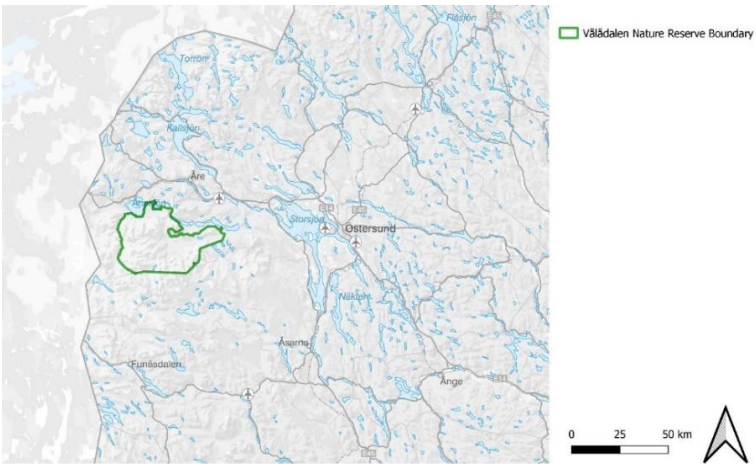
The research design was divided into two components, a geotag distribution analysis and a human impact analysis. The distribution analysis was performed to gain an understanding of the relationship between geotag distribution and existing management plans and if it can be used to evaluate visitor behaviour patterns. The human impact analysis was performed to determine if there was a correlation between geotag hotspots and human impact that could be identified using aerial imagery.

## 2.1 Study Area

Vålådalen Nature Reserve lies in Jämtland, 30km south-west of the ski-town of Åre and 100km to the west of the city of Östersund (Figure 2). It has an area of 1211km$^2$ and is composed of several different landscape types; primarily old-growth forests, alpine areas and wetlands, and is an area of habitat for multiple endangered species (Länsstyrelsen Jämtland, n.d.).

**Figure 2**

*Vålådalen Nature Reserve location in Western Jämtland*




*Note:* Map of Western Jämtland outlining Vålådalen Nature Reserve**.** Map created by Author. Map data: [Lantmäteriet, Naturvårdsverket] (2026). Produced using QGIS 3.22.2. QGIS Association. http://qgis.org

The area is utilized for tourism in both summer and winter, with an extensive trail network and 4 seasonally operated STF Cabins within the boundary of the reserve. The area is managed by Länsstyrelsen Jämtland and is a grazing

area for reindeer from both the Handöl and Tossåssen Sámi Villages (Länsstyrelsen Jämtland, n.d.). Multiple land use conflicts have occurred within the reserve, most recently between the Sámi RHC's and the Svenska Turistföreningen (STF) in 2023 regarding the renewal of leases on STF cabins and mountain stations located in the area (Svenska Turistföreningen, 2026). Land use conflict has also occurred in recent years at the icon sites of Östra Blanktjärnarna, Issjödalen and Pyramiderna, where there are now restricted zones in place (from 2019), due to the damage caused and potential for further damage to occur due to tourism (Länsstyrelsen Jämtland, n.d. -a).

Vålådalen Nature Reserve was chosen as the study area for this report due to its visitor numbers, extensive infrastructure, detailed zoning plan, recorded land use conflicts and the large quantity of geotag data available.

## 2.2 Data Collection and Initial Processing

Topographic maps, activity zoning maps, height data, trail datasets and aerial vegetation surveys were downloaded from open data sources online (Table 1) and imported into QGIS (QGIS 3.22.2.). Visualization of the layers was performed using QGIS. Instagram and Flickr were both used as sources for geotag data, with differences between the type of data they provide and the users they are aimed at.

Instagram is a social media application which only allows photographs to be uploaded and geotagged on a phone application to available "sites", which are geographical locations that are pre-determined by the application, regardless of the exact geographical position that the photograph was recorded in. In 2025 Instagram has over 3 billion active users and over 60% of them were between the ages of 18-35 (Jaimes, 2025), (Pogačar & Žižek, 2020). Flickr is available as both a website and a phone application and was originally aimed at amateur photographers. The user who uploads the photograph can either geotag the exact location a photograph was taken by manually choosing the location, or by using the geographical metadata stored by their mobile telephone or camera. In 2022 Flickr had 60 million active users (Leppämäki et al., 2025), (Pogačar & Žižek).

Both Instagram and Flickr were chosen as sources of geotags based on the availability of data access, the quantity of data available for analysis, and their well-established use in previous studies (Wilkins et al., 2020).

**Table 1**

*Data sources, dataset names, layer types and retrieval dates for layers imported into QGIS*

| Project Layer Name | Layer Type | Source | Name of Dataset at Source | Date Retrieved |
|---|---|---|---|---|
| Topographic Map | WMS | Läntmateriet (Lantmäteriet, n.d.) | Topografisk Webbkartan | 2026/02/16 |
| Flickr Geotag Data | Delimited Text File | Manually collected from Flickr Website | N/A | 2026/03/30 |
| Instagram Geotag Data | Delimited Text File | Manually collected from Instagram Application | N/A | 2026/04/16 |
| Nature Reserve Boundaries | ESRI Shapefile | Naturvårdsverket (Naturvårdsverket, n.d.) | Skyddade områden: naturvårdsområden | 2026/03/08 |
| Summer Trails | String (GPKG) | SLU (SLU, n.d.) | Topographi 10 | 2026/03/30 |
| Winter Trails | String (GPKG) | SLU (SLU, n.d.) | Topographi 10 | 2026/03/30 |
| Orthophoto RGB | GeoTiff | SLU (SLU, n.d.) | Ortofoto RGB 0.5 m 2002-2018,<br>Ortofoto RGBI latest 0.16/0.4 m | 2026/02/16<br>2026/05/03 |
| Vålådalen Zone Map | GeoTiff | Länsstyrelsen Jämtlands Län (Länsstyrelsen Jämtland, n.d. b) | LstZ Zonering & restriktionsområden Vålådalen | 2026/04/28 |
| Hydrography | Shape (GPKG) | SLU (SLU, n.d.) | Fastighetskartan Hydrografi latest | 2026/04/16 |

### 2.2.1 Flickr Geotag Data Collection

The study area was divided into search boxes using the QGIS "Create grid" function, with each grid box corresponding to the effective area of Flickr's Map search tool. Each box was manually searched using Flickr's website map tool and every photograph identified within the study area was assigned an ID, and the latitude ($^0$), longitude ($^0$), and creation date (YYYY-MM-DD) were recorded manually using Microsoft Excel.

The data was formatted as a delimited text file and imported into QGIS. Photographs in which the date and both coordinates were identical, were removed from the dataset to reduce bias towards a single operator who took multiple photos in the same place on the same day. In QGIS the data was filtered by date (month) to create a dataset of geotags that were recorded in bare ground season, between May and October, months primarily without snow on the ground, and a dataset of geotags that were recorded in snow season, between November and April, months primarily with snow on the ground. The months that belong to each period were determined using historical data available on SMHI's Season Arrival Map service (SMHI, n.d.). The data from both dataset categories was displayed as points styled with alternate colours on a topographic map of the Vålådalen Nature Reserve Area.

### 2.2.2 Instagram Geotag Data Collection

The Instagram phone application was used to identify the number of Instagram photos that had been attributed to each pre-determined Instagram geotag location site within the study area. Only photos tagged on the "feed" function were evaluated. 22 geotag locations were found within the study area and for each location the latitude($^0$), longitude($^0$) and the number of geotagged media attributed to that location between January 2015 and December 2025 were manually recorded using Microsoft Excel.

The data was formatted as a delimited text file and imported into QGIS, then displayed as points on a topographic map of the Vålådalen Nature Reserve Area. Each location datapoint was visually styled to have a size proportional to the number of geotags identified at that location.

## 2.3 Distribution Analysis

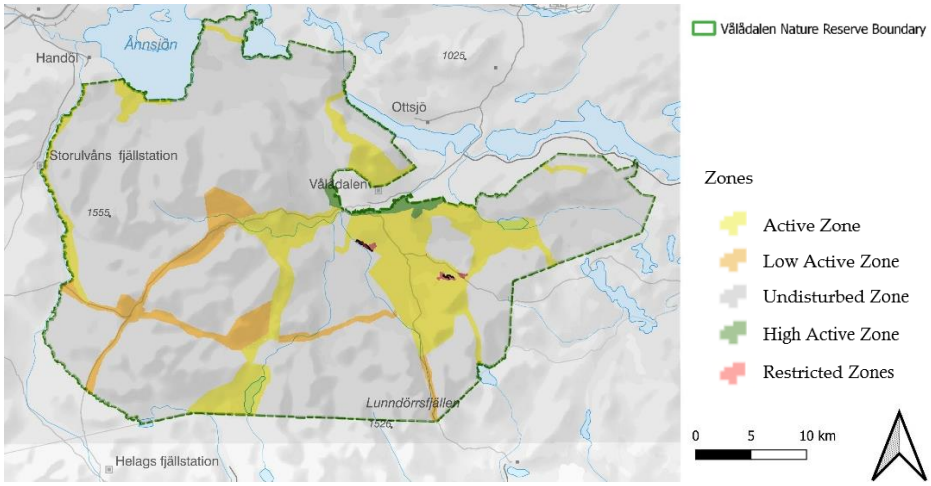
The first distribution analysis technique is a comparison of geotag location distribution against reserve activity zones, which have been created by Länsstyrelsen Jämtland as part of the ongoing management of the nature reserve (2.3.1. Zone Geotag Density Analysis). This technique quantifies the alignment between the current reserve zoning system and visitor density identified in the geotag datasets. The second distribution analysis technique uses the Flickr dataset and compares the geotag location distribution with the trail network of the reserve, (2.3.2. Analysis of geotag proximity to trails). This technique aims to establish if seasonal information about trail usage can be concluded from the data.

### 2.3.1 Zone Geotag Density Analysis

The Vålådalen Activity Zoning Map (Figure 3) was imported from Länsstyrelsen, and the decision document "Reservatsbeslut Vålådalen" was used to determine and record the following characteristics for each of the 5 zones that exist in the reserve; The relative frequency of visitation, values, infrastructure, and restrictions. (Länsstyrelsen Jämtlands län, 2023).

**Figure 3**

*Vålådalen Nature Reserve Activity Zoning Map*




*Note:* Vålådalen Nature Reserve map with activity zone areas overlayed. Map created by Author. Map data: [Lantmäteriet, Länsstyrelsen Jämtland] (2026). Produced using QGIS 3.22.2. QGIS Association. http://qgis.org

The zoning map was imported into QGIS as a raster layer type and the processing tool, "Classify by Table" was used to assign a separate value to each zone. The layer was converted into a vector layer using the tool, "Polygonize (Raster to Vector)", and the tool, "Split Vector Layer" was used to create a separate shapefile layer for each value and therefore each zone.

The Flickr geotag dataset was compared to each zone using the tool, "Count points in a polygon", where the geotags were the points and the zones were the polygons respectively. The results were recorded as the number of unique points within each polygon (zone). The Instagram geotag dataset was also compared to the zones using the tool, "Count points in a polygon", however as the Instagram dataset is visualized as one geodata point with the number of geotags represented by the value, the zone each point is located in was recorded and Microsoft Excel was used to calculate the number of geotags in each zone using the raw value data collected from Instagram.

The "Field Calculator" was used to determine the area (**A**) of each zone in km². The raw geotag count (**Gn**) in each zone for each dataset and the density (**Gd**) of the geotags in Geotags per km² per zone, was calculated using Equation 1 in Microsoft Excel.

$$\mathbf{Gd = Gn/A} \qquad \mathbf{(1)}$$

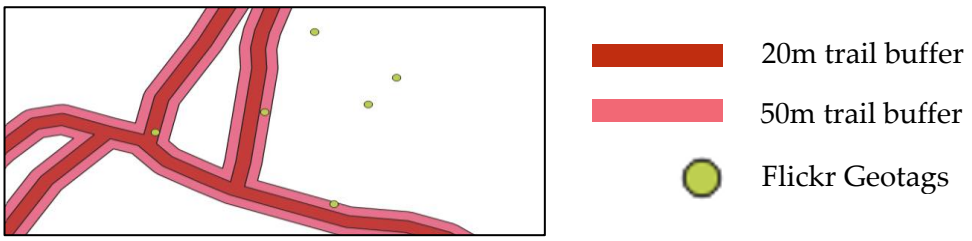
The raw geotag count per zone and the geotag density per zone were displayed as bar charts using Microsoft Excel.

### 2.3.2 Analysis of Geotag Proximity to Trails

The summer and winter trail networks were imported into QGIS and the geoprocessing tool "buffer" was used to create a 20m buffer zone and a 50m buffer zone (Figure 4) for the entire network of marked trails within the reserve. The 20m buffer was chosen to cover visitors who are directly using the trails, while making an allowance for small inaccuracies in geotag and trail datasets. The 50m buffer was chosen to cover short departures from the trail and movement taking place on social trails alongside the main trail, such as visiting viewpoints.

**Figure 4**

*20m and 50m buffer zone illustration with Flickr geotag datapoints*




*Note:* Cut-out of buffered trail section showing 20m buffer, 50m buffer and Flickr Geotag Datapoints. Image created by Author (2026). Produced using QGIS 3.22.2. QGIS Association. http://qgis.org

The Flickr datasets were separated for snow vs bare ground season (2.2.1 Flickr Geotag Data Collection), and the trail network buffers were processed using the "The Join Attributes by Location (Spatial Join)" to determine the number of geotags overlapping with each trail buffer with respect to season. The geometric relationship required to accept geotags as being within the buffers was set to "intersect", whereby if the geotag datapoint touched the buffer in any way it was included in the count. Intersect was chosen as it would not exclude data that was only partially within the buffer zone.

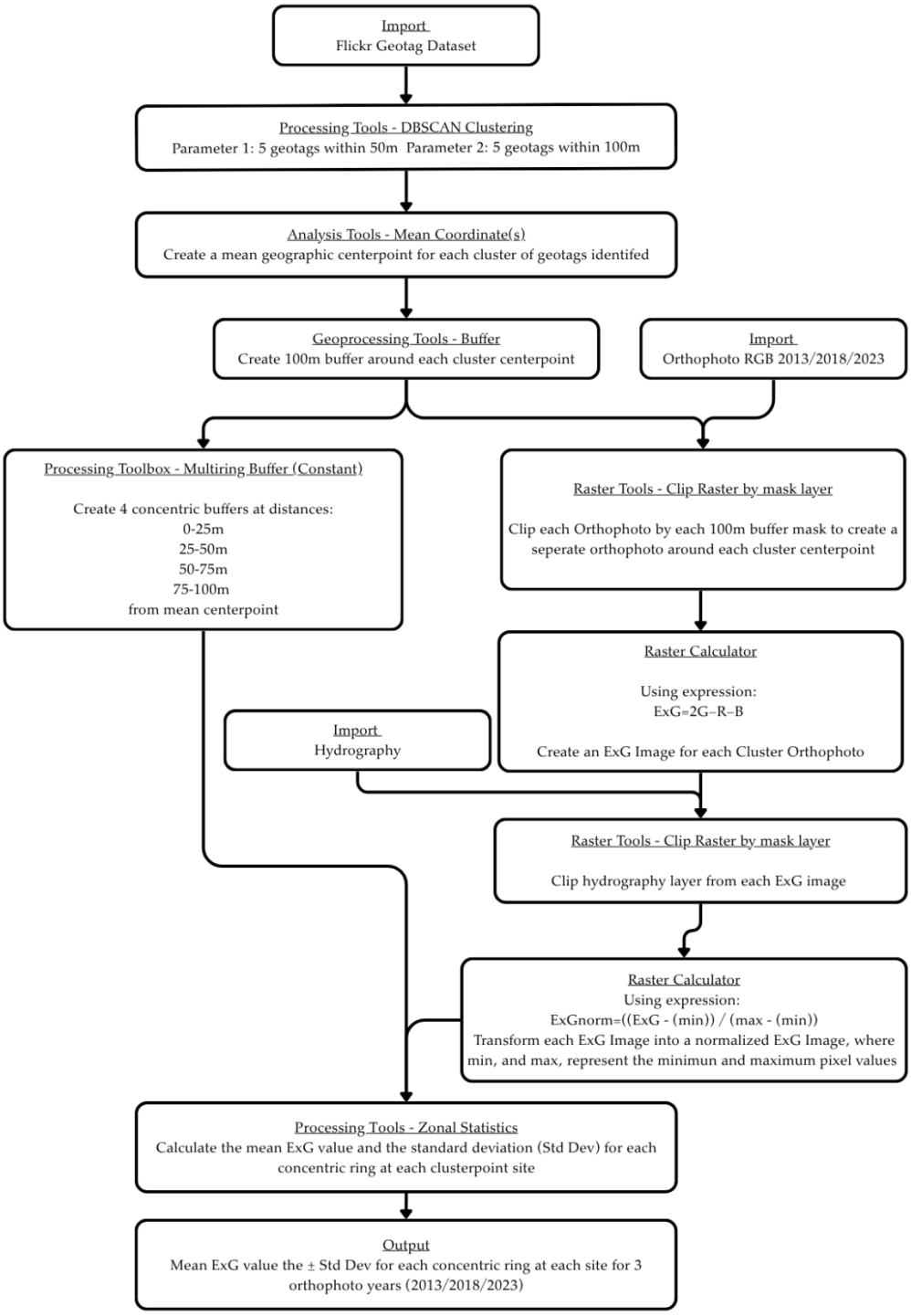
The data was recorded in Microsoft Excel and displayed as bar charts showing the number of points in each trail buffer separated by snow vs bare-ground season. The Instagram geotag data was unsuitable for this analysis.

## 2.4 Human Impact Analysis Method

To determine if there is a relationship between concentrated geotagging with human impact on the local environment, the following methodology used analysis of RGB orthophotos to calculate the mean Excess Green Index (ExG) of concentric circle nested buffers at geotag cluster points. As human impact such as trampling and vegetation damages decreases, the mean ExG values should increase (Woebbecke et al., 1995). Using the Flickr Geotag dataset cluster points were identified using the parameters, "5 geotags within 50m" and "5 geotags within 100m" in the processing tool, "DBSCAN Clustering". The study areas were determined by creating a geographic mean centerpoint for each cluster point using the analysis tool' "Mean Coordinates".

**Figure 5**

*Impact Analysis QGIS Methodology as a flowchart*



The full impact analysis workflow methodology in QGIS is shown in Figure 5. RGB orthophotos was chosen as there was increased availability for a larger range of dates, allowing for comparison between more orthophoto-dates in comparison to using IR orthophotos. Orthophotos were downloaded for 2013, 2018 and 2023 respectively. The tool "Raster Calculator" was used to create an **ExG** (Excess Green Index Image) and extract the green spectral light from the RGB Orthophotos. This allowed vegetation to be distinguished from the surrounding environment using Equation 2, where **G**, **R** and **B** are the Green, Red and Blue layers in the imported orthophoto layers respectively.

$$\mathbf{ExG = 2G - R - B} \qquad \mathbf{(2)}$$

An ExG method was chosen as it has been shown in research to outperform other indices that can be applied to RGB images to discriminate greenery (Larrinaga & Brotóns, 2018), (Srinivasan, 2024). Areas of water were masked from the orthophotos using a hydrology layer, as water can create background disturbance and influence the mean ExG values of an area (Yuan et al., 2022). The tool, "Raster Calculator" was used to normalize the ExG images (**ExG**), so every pixel would have a value between 0-1, using Equation 3.

$$\mathbf{ExGnorm = ((ExG - (min)) / (max - (min))} \qquad \mathbf{(3)}$$

In Equation 3, **ExGnorm** is the normalized ExG image, and **max** and **min**, represent the maximum and minimum pixel values present in that specific ExG, available in the properties of each image on QGIS. Normalizing the image allows for comparison to take place between mean ExG values of different analysis. Comparison only took place between areas on orthophotos taken at the same time, as environmental variations such as sun angle, cloud cover and seasonal green coverage will occur between orthophotos taken at different dates.

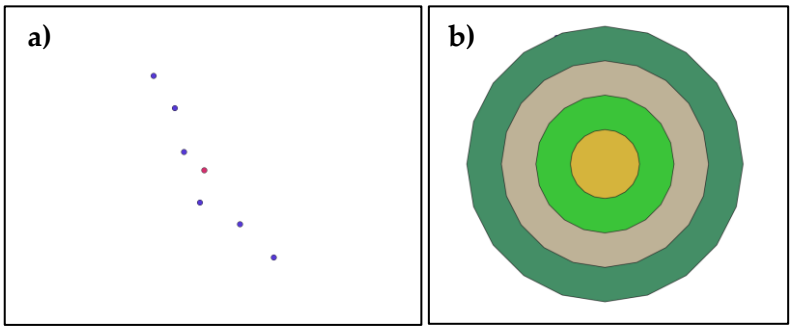
Nested buffers of concentric circle shape were used to create 4 distinct distances from the mean geographic centerpoint of the identified clusters (Figure 6). These were at 0-25m, 25-50m, 50-75m and 75-100m respectively.

Using concentric nested buffers is recommended when the goal is assessing greenspace at varying distances independently (Sakhvidi et al., 2025).

The "Zonal Statistics" tool was used to determine a mean ExG value and Standard Deviation (Std Dev), for each concentric buffer, for each orthophoto year, at each location identified by the parameters.

**Figure 6**

*Illustration of creation of mean geographic centerpoint and nested buffers*




*Note:* a) Creation of a mean geographic centerpoint (red) from clustered geotags (blue). b) Nested buffers of concentric circles. Diagram created by Author. Map data: [Flickr Geotag Dataset] (2026). Produced using QGIS 3.22.2. QGIS Association. http://qgis.org

In Microsoft Excel the mean ExG values were plotted as clustered column charts, separated by location with respect to orthophoto date. Additionally the Standard Deviation determined by QGIS was plotted for each value.

To determine if there was a relationship between distance from the cluster center points and the mean ExG values, the frequency of an increase in the mean ExG value for between the 0-25m and the 25-50m, the 25-50m and the 50-75m, and the 50-75m and the 75-100m buffers were calculated as a percentage for both of the cluster point parameters, "5 geotags within 50m" and "5 geotags within 100m".

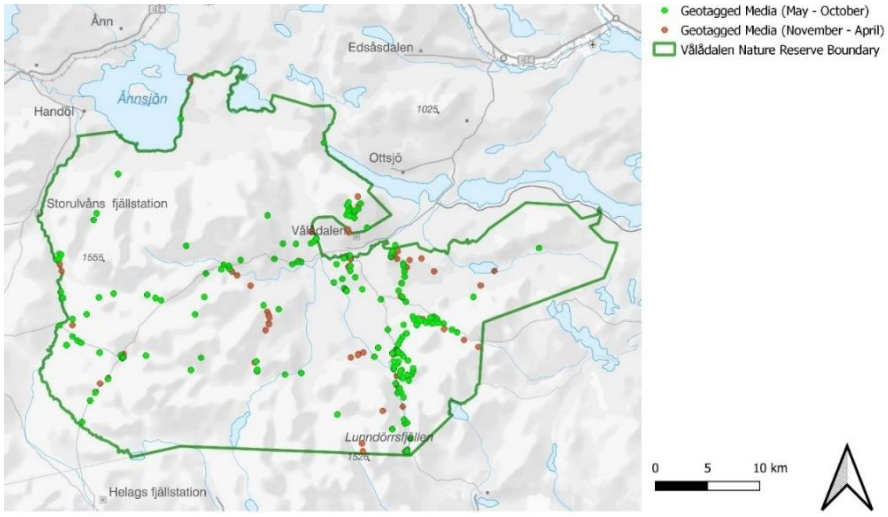
The frequency of increase directly between the 25-50m buffer (closest to the centerpoint) and 75-100m buffer (furthest from the centerpoint) was also calculated as a percentage for the parameters combined.

# 3 Results

A total of 412 geotags were collected from Flickr. 296 were recorded in the bare ground season of May to October and 116 were recorded in the snow cover season of November to April. The geotags are shown plotted as points (Figure 7) with respect to the season. The images were taken between February 2005 and August 2023.

**Figure 7**

*Map visualizing distribution of Flickr data*

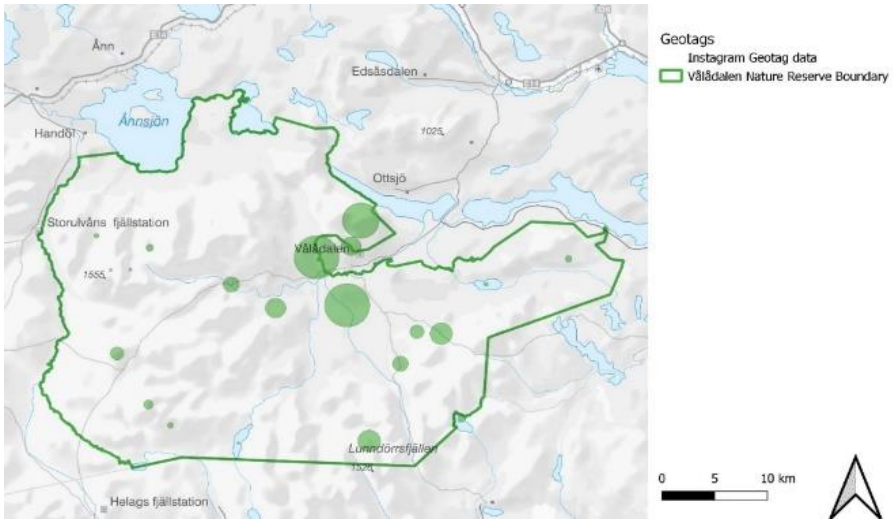



*Note:* Map of Vålådalen Nature Reserve with a) Instagram Geotags sized by value b) Flickr geotags locations split by season. Map created by Author. Map data: [Lantmäteriet, Naturvårdsverket, Flickr, Instagram] (2026). Produced using QGIS 3.22.2. QGIS Association. http://qgis.org

5064 geotags were collected from Instagram from 22 different sites pre-determined by the application (See Appendix A for full results list). Plotting them as points sized by value (Figure 8) showed concentrated areas of data surrounding the village Vålådalen itself, and 3 sites with considerably higher values, at "Ottfjället 1265 m. ö. h.", "Blanktjärn" and "Nulltjärnarna", which had 816, 1193, and 1185 geotags respectively.

**Figure 8**

*Map visualizing distribution of Instagram geotag data*




*Note:* Map of Vålådalen Nature Reserve with a) Instagram Geotags sized by value b) Flickr geotags locations split by season. Map created by Author. Map data: [Lantmäteriet, Naturvårdsverket, Flickr, Instagram] (2026). Produced using QGIS 3.22.2. QGIS Association. http://qgis.org

## 3.1 Geotag Count and Density by Zone

Table 2 aggregates the results of the review of the document "Reservatsbeslut Vålådalen". The document details the different visitation frequencies expected in each zone as part of the management of Vålådalen Nature Reserve, and the infrastructure and restrictions in place to manage visitors in those areas. The zones, "Active" and "High Active" expect to receive the highest number of visitors, and the most infrastructure to accommodate this and the least restrictions, while the zone, "Undisturbed" expects the least visitors, has no formal infrastructure and the most restrictions.

The restricted zones are parts of the active zone, expecting high visitation numbers with extensive infrastructure in place to accommodate this, but with strict site specific restrictions in place. In the areas of Östra Blanktjärnarna, Issjödalen and Pyramiderna visitors are restricted to staying on designated trails, while cycling, horseback riding, camping, making fires and entering water bodies are forbidden.

## Table 2

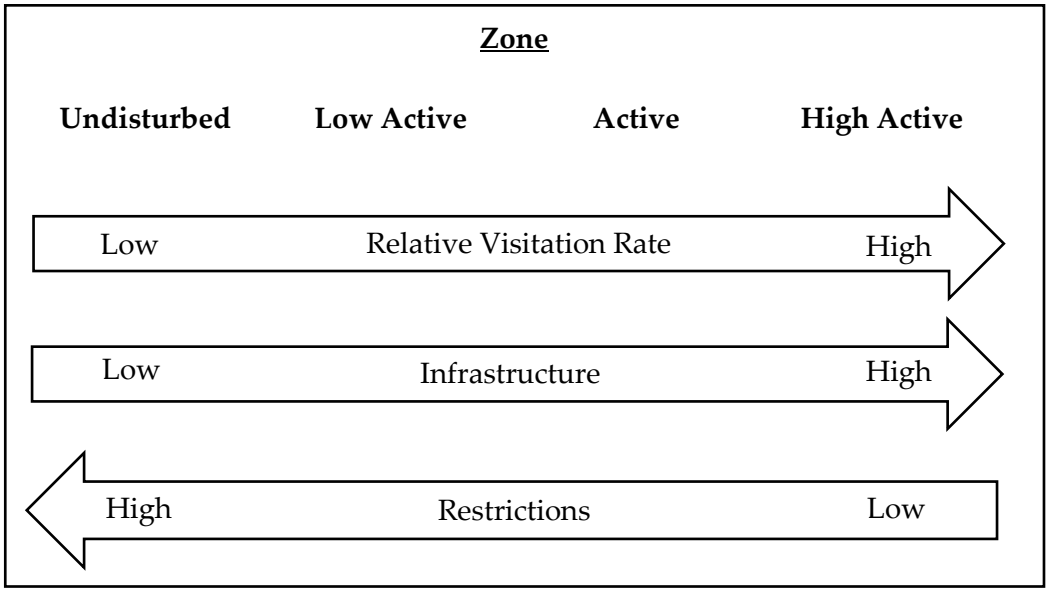
*Types of zones and activity level interpretation*

| Zone Name | Relative frequency of visitation | Values | Infrastructure | Restrictions |
|---|---|---|---|---|
| High Active | For channelling the largest number of visitors | Natural, cultural and recreational | Clear information, well maintained trails, accessible trails, toilets, rest areas, campfire sites | Fires restricted except for in fire pits, collecting firewood forbidden, snowmobiling only permitted on trail. |
| Active | Well visited, frequent use | Natural, cultural and recreational, wilderness | Trails, rest areas, overnight cabins, fireplaces, dry toilets | Organised recreation requires permits, cycling only on designated trails, only small branches and twigs can be collected as firewood, snowmobiling only permitted on trail. |
| Low Active | Lower visitation, recommended not to visit between April 20th-June 30th | Natural, cultural, recreational, silence, wilderness | Lower standard trails, less accessible, overnight cabins | No organised recreation, no cycling, no snowmobile trails, only small branches and twigs can be collected as firewood. |
| Undisturbed | Very low visitation, recommended not to visit between April 20th-June 30th | Natural, cultural, recreational, silence, wilderness, reindeer herding | No trails, no structures | No organised recreation, no horseback riding, no cycling, no snowmobile trails, only small branches and twigs can be collected as firewood. |
| Restricted (Within Active) | As per "Active" | Natural, cultural and recreational, unique landscape values | Clear information, well maintained trails | Not permitted to leave the trails, no cycling, no camping, no fires, no horseback riding, no entering water bodies. |

The relationships between the zones and, visitation frequency, infrastructure, and restrictions are shown in Figure 9.

**Figure 9**

*Diagram showing zones by visitation rates, restrictions and infrastructure.*




The calculated area of each zone, number of geotags recorded, and the geotags density in geotags per km² (Gn/ km²) per zone are displayed in Table 3 (Flickr Geotag Dataset) and Table 4 (Instagram Geotag Dataset).

**Table 3**

*Zone category raw geotag count and density analysis data for Flickr Data*

| **Zone** | **Area A (km²)** | **Geotag Count Gn** | **Geotag Density Gd (Gn/km²)** |
|---|---|---|---|
| High Active | 8 | 25 | 3,13 |
| Active | 198 | 221 | 1,12 |
| Low Active | 59 | 73 | 1,24 |
| Undisturbed | 940 | 69 | 0,07 |
| Restricted | 1,9 | 24 | 12,63 |

**Table 4**

*Zone category raw geotag count and density analysis data for Instagram Data*

| Zone | Area A ($km^2$) | Geotag Count Gn | Geotag Density Gd ($Gn/km^2$) |
|---|---|---|---|
| High Active | 8 | 1185 | 148,13 |
| Active | 198 | 1537 | 7,76 |
| Low Active | 59 | 111 | 1,88 |
| Undisturbed | 940 | 725 | 0,77 |
| Restricted | 1,9 | 1506 | 792,63 |

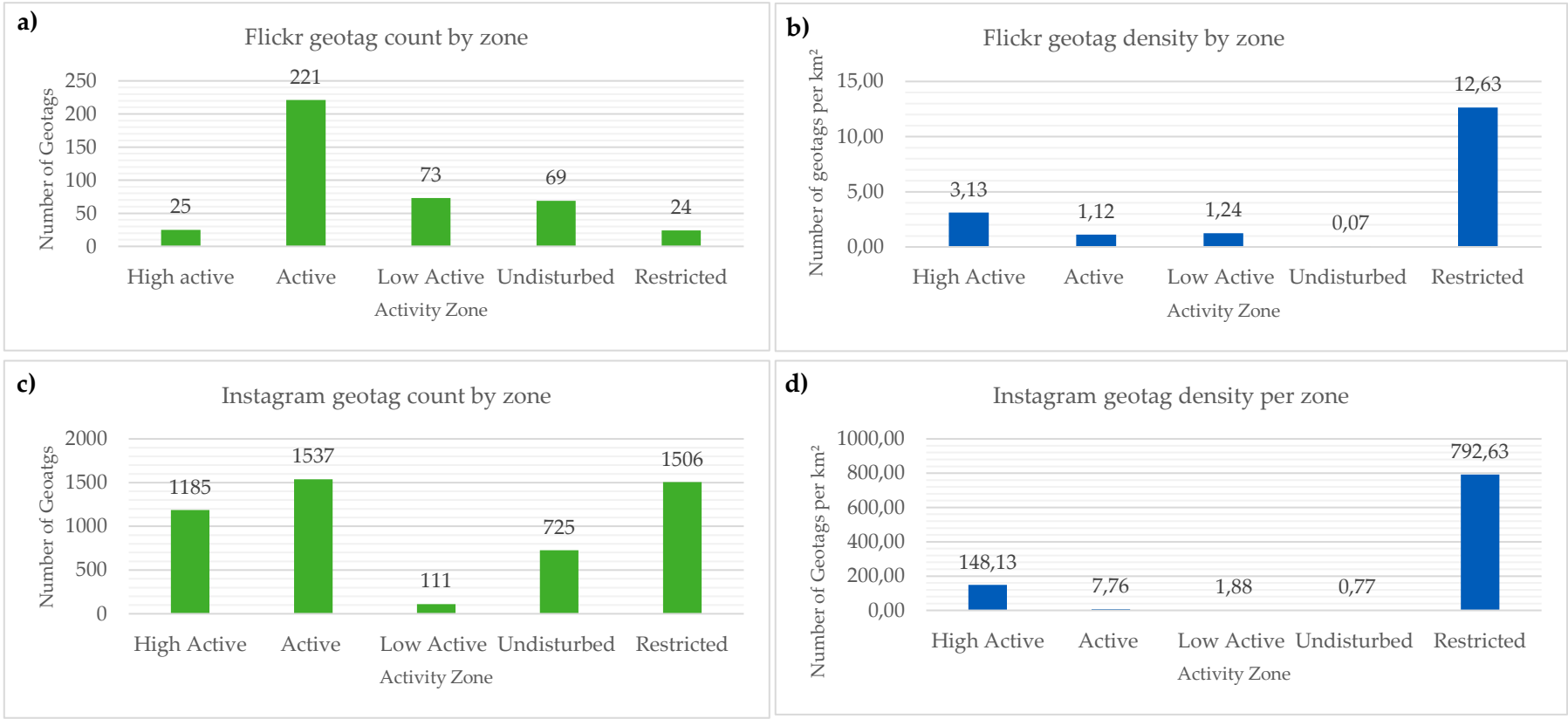
The zone with the largest combined area in the reserve is the undisturbed zone, with an area of 940 $km^2$, representing 78% of the total area of the reserve. The smallest zone in the restricted zone with an area of 1,9 $km^2$, making up 0,002% of the reserve.

Figure 8 displays the geotag count per zone and geotags density plotted per zone for Flickr (Figure 10 - a) and b)) and Instagram (Figure 10 – c) and d)) respectively. The geotag count peaked in the Active zone for both datasets, with 221 Flickr geotags and 1527 Instagram geotags. The lowest number of geotags was recorded in the restricted zone for the Flickr dataset, with only 24 geotagged photographs recorded, while in contrast, the restricted zone in the Instagram dataset had the second highest geotag count at 1506. The lowest number of geotags recorded in the Instagram dataset was in the Low Active zone, at 111.

For both the datasets the geotag density peaked in the Restricted zone, followed by the High Active, the Active, then the Low Active, with the Undisturbed zone receiving the lowest density of geotags in both datasets.

**Figure 10**

*Instagram and Flickr geotag count and geotag density by zone*



## 3.2 Trail Network and Geotag Proximity

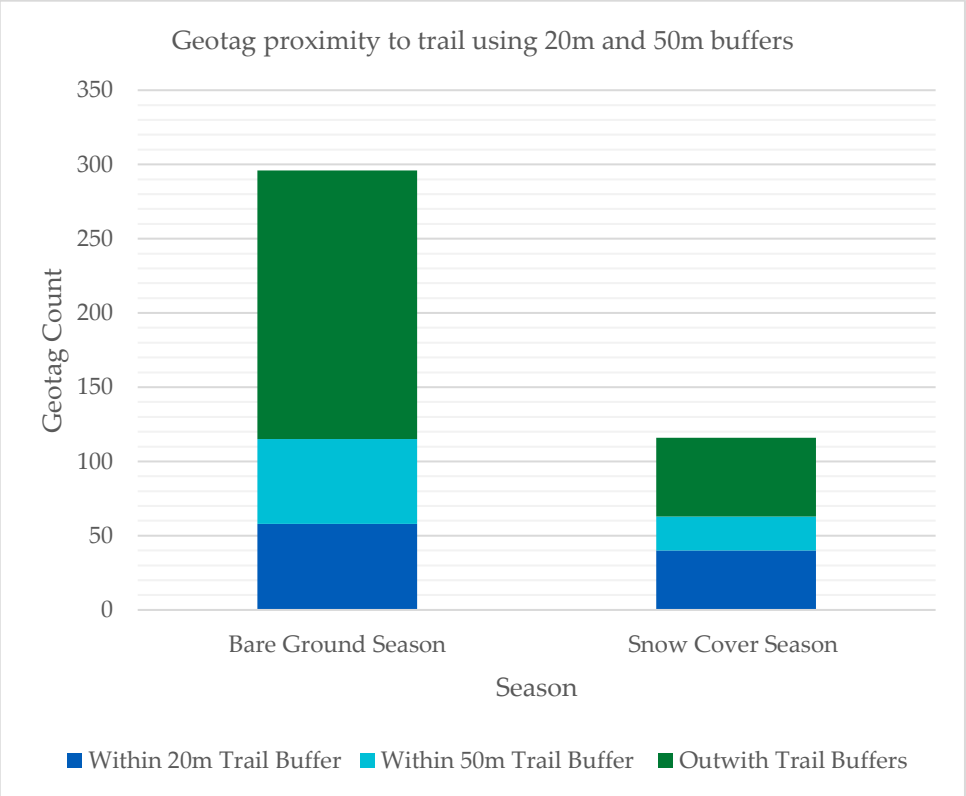
98 of 412 (23%) Flickr Geotags were determined to be on or within 20m of a marked trail, and 178 of 412 (43%) were determined to be within 50m of a marked trail. Figure 9 provides a breakdown of tags within the 20m trail buffer, the 50m trail buffer, and those out with either trail buffer, with respect to season.

19% of geotags recorded in bare ground season were determined to be on or within 20m of a trail, while 39% were within 50m of a trail.

In the snow cover season 23% of geotags were determined to be on or within 20m of a trail, and 54% were within 50m of a trail.

**Figure 11**

*Graphical representation of Flickr Geotags within and out with trail buffers*



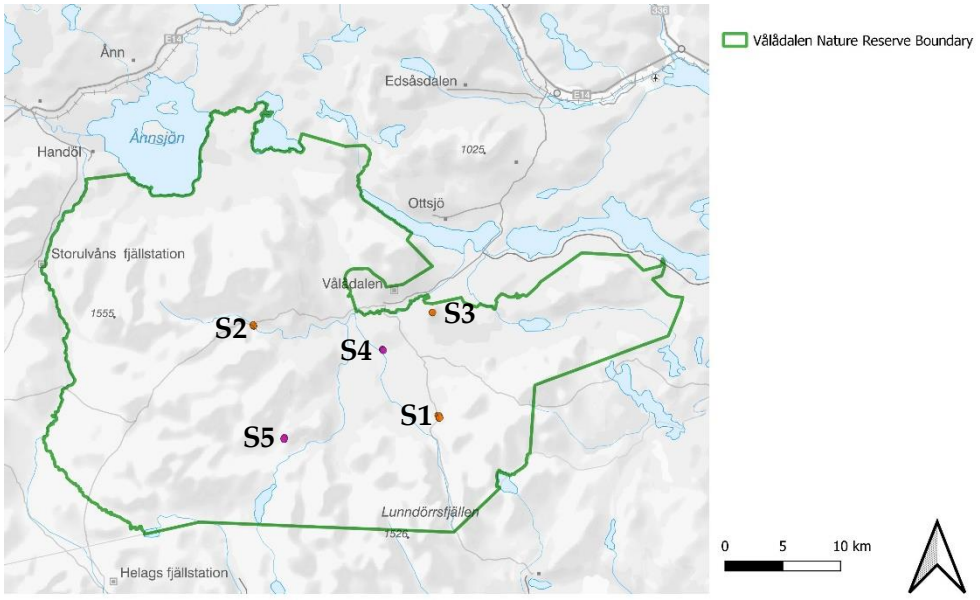
## 3.3 Human Impact Results

The parameter, “5 or more geotags within 50m” identified 3 cluster sites, labelled as sites S1, S2 and S3 (Figure 11) and the parameter, “5 or more geotags within 100m” identified 2 cluster sites, labelled as sites S4 and S5 (Figure 11), see Appendix B for full site list.

All 5 sites corresponded closely with sites with infrastructure within the reserve. S1, S2 and S5 occur at Lunndörrsstugan, Stensdalsstugorna and Vålåstugan, each site is the location of an STF Fjällstuga (Mountain Cabin). S3 is at the site of Middagsvalen, a viewpoint, and S4 is at the site of Blanktjärn, a popular day trip destination, and one of the restricted zones within the reserve.

**Figure 12**

*Locations that met the parameters for Orthophoto analysis and associated zone*



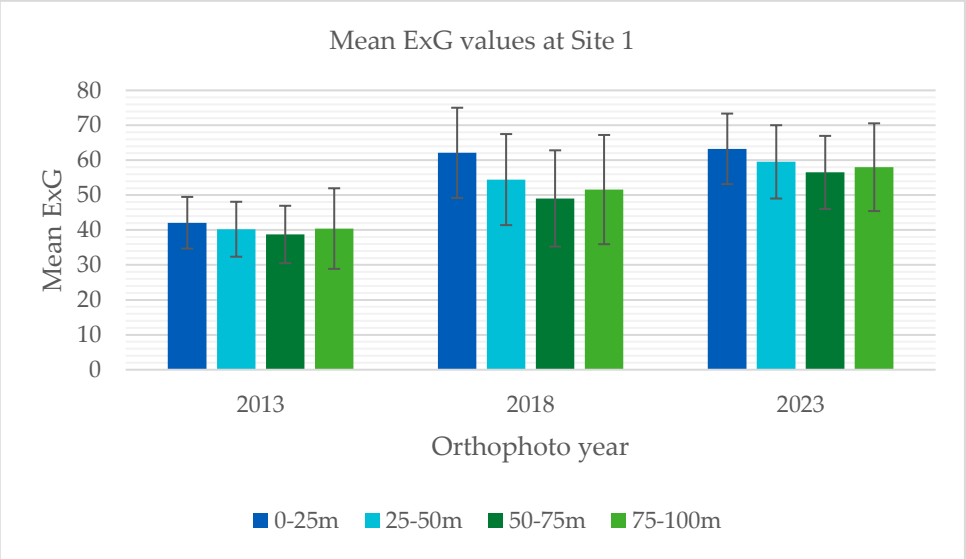

*Note:* Map of Vålådalen Nature Reserve with locations of geotag clusters and associated ID numbers. Map created by Author. Map data: [Lantmäteriet, Naturvårdsverket, Flickr ] (2026). Produced using QGIS 3.22.2. QGIS Association. http://qgis.org

The results at each site (Figure 12, 13, 14, 15, 16), show the Mean ExG in 4 buffer zones moving out from the mean center point of the cluster, across 3 years of sampling. The error bars represent the Standard Deviation, showing the variations of the values of each zone.

At Site 1 (Figure 13), the results show a similar gradient across all 3 years, with the highest ExG consistently occurring in the 0-25m zone closest to the cluster center point, and the lowest ExG occurring in the 50-75m zone from the centerpoint.

**Figure 13**

*ExG Mean for each concentric buffer at Site 1, across 2013, 2018 and 2023*




At Site 2 (Figure 14), the mean ExG values for 2013 and 2018 are greatest in the 0-25m buffer. In 2013 the values decrease in each buffer as the distance from the centerpoint increases. In 2018 the values decrease to the 50-75m buffer but there is a slight increase in the mean ExG value in the 75-100m buffer. In 2023 there is the least variation across the sampling years, with the 25-50m buffer having the lowest value and the 75-100m buffer having the highest value.

**Figure 14**

*ExG Mean for each concentric buffer at Site 2, across 2013, 2018 and 2023*

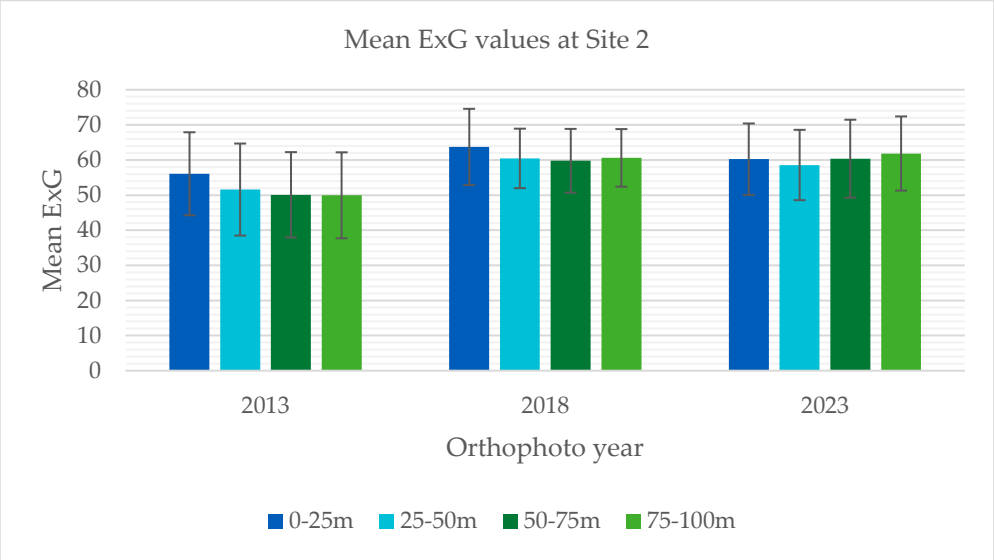
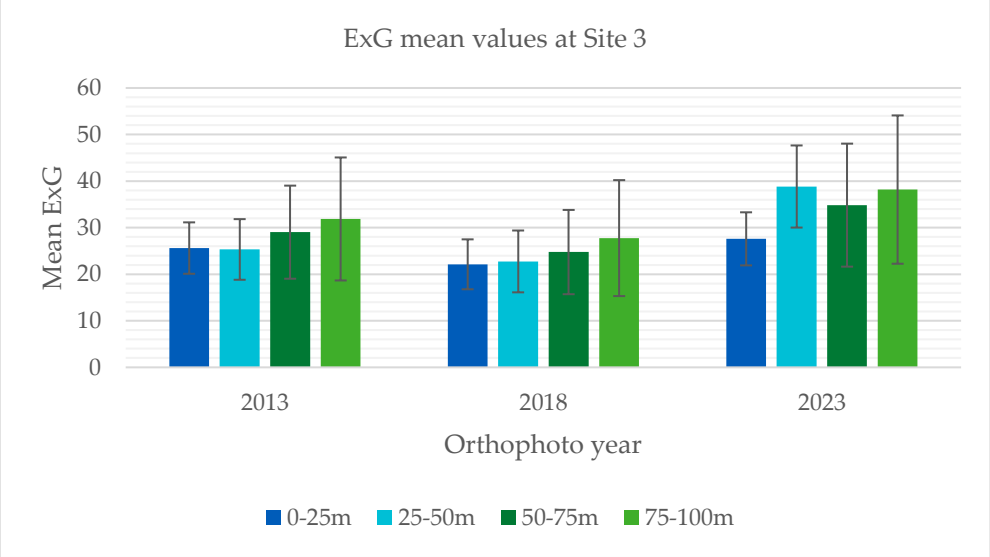



**Figure 15**

*ExG Mean for each concentric buffer at Site 3, across 2013, 2018 and 2023*

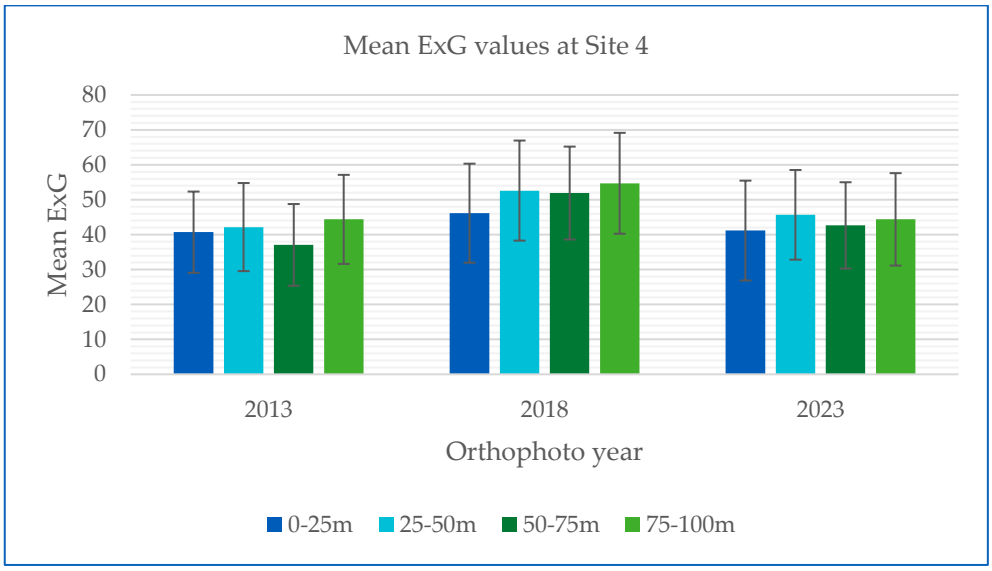


At Site 3, (Figure 15) the 2013 and 2018 samples follow a similar gradient, with the lowest mean ExG occurring in the 25-50m buffer and the 0-25m buffer respectively, with an increase in mean ExG in the 50-75m buffer and the highest ExG occurring in the 75-100m buffer. In the 2023 sample the lowest mean ExG value occurs in the 0-25m buffer and the highest in the 25-50m buffer.

At Site 4 (Figure 16) there is a similar trend across all 3 years. In 2018 and 2023 the mean ExG is lowest in the 0-25m buffer, and across all 3 sampling years the mean ExG is increases in the 25-50m buffer, decreases to the 50-75m buffer and then increases again to the 75-100m buffer. The 2023 image shows a greater range of values between the means.

**Figure 16**

*ExG Mean for each concentric buffer at Site 4, across 2013, 2018 and 2023*




At Site 5 (Figure 17), there are different patterns across all 3 sampling years. In 2013 and 2023 the lowest mean ExG occurs in the 0-25m buffer. In 2013 and 2018 the highest value occurs in the 50-75m buffer. In 2023 the values increase in each sampling ring as the distance from the center point increases.

**Figure 17**

*ExG Mean for each concentric buffer at Site 5, across 2013, 2018 and 2023*

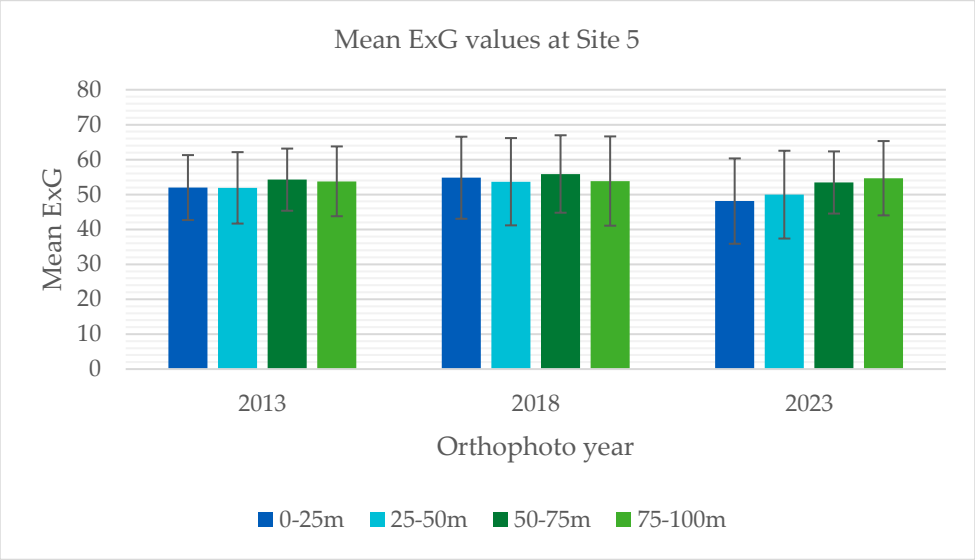



Table 5 shows the frequency of the increase in the mean ExG values between each set of buffers as a percentage. Between the 0-25m buffers, 25-50m and 50-75m buffers the frequency of an increase in the mean ExG value occurred on 40% or fewer occasions. Between the two outermost buffers, 50-75m and 75-100m buffers, the frequency of an increase in mean ExG value was 80% where there was a cluster of 5 or more geotags within 100m and 88% where there was a cluster of 5 of more geotags within 100m.

**Table 5**

*Frequency of increase in mean ExG between buffers across all sampling dates*

| | **Frequency of increase of mean ExG** | | |
|---|---|---|---|
| **Parameter** | **0-25m/25-50m** | **25-50m/50-75m** | **50-75/75-100m** |
| **5 or more geotags in 50m** | 22% | 33% | 88% |
| **5 or more geotags in 100m** | 40% | 40% | 80% |

When the mean ExG values 0-25m buffer were compared directly to the corresponding values taken in the 75-100m buffer it was determined that there was an increase in mean ExG value on 8 of 15 sampling occasions, corresponding to an increase in mean ExG value on 53% of occasions.

# 4 Discussion

The purpose of this study was to investigate whether crowdsourced geotagged social media data can be used to determine visitor behaviour in Swedish Nature Reserves and considers if there is a relationship between the intensity of geotag activity and human impact within these protected areas.

## 4.1 Discussion of Results

Overall, the results suggest that geotagged data from Flickr and Instagram can provide information regarding visitor behaviour and visitor distribution patterns within Vålådalen Nature Reserve. However, the relationship between geotag cluster points of activity and ecological impact was less clear.

### 4.1.1 Geotag Distribution and Zoning

The geotag distribution and zoning analysis demonstrate that the Flickr and the Instagram datasets have similar patterns of distribution throughout Vålådalen Nature Reserve.

The distribution of the raw geotag count does vary between the datasets, however the highest raw geotag counts exists in the Active Zone of both datasets. The geotag densities for both datasets show the same pattern of having the highest density in the restricted zone, followed by the high active zone, active zone, low active zone and the lowest density occurs in the undisturbed zone.

This visitor density distribution pattern aligns with the relative visitation rate that Länsstyrelsen Jämtland has described in their management decision document, "Reservatsbeslut Vålådalen" for the nature reserve, which is visualized in Figure 9. This suggests that the highest visitor activity levels take place in the zones where the infrastructure such as trails, toilets and cabins are provided to manage and accommodate high visitor numbers.

The restricted zone clearly has the highest density of geotags in both datasets, whilst also having the lowest area. This suggests highly concentrated visitation in these areas of the reserve. Page 35 of "Reservatsbeslut Vålådalen" (Länsstyrelsen Jämtlands län, 2023) describes the extensive damage at the site of Blanktjärn, one of the restricted areas in the reserve. The Instagram dataset (Appendix A), also identifies Blänktjärn as having the highest geotag count within the reserve.

These findings using both the Flickr and Instagram Geotag datasets for Vålådalen Nature Reserve support previous studies that have concluded that geotagged social media data can effectively identify visitor hotspots and spatial patterns of recreation use (Geneletti & Orsi, 2013), (Levin et al., 2017), (Wilkins et al., 2020). This relationship between the geotag patterns and the reserve zoning system suggests that social media data may provide a useful source of visitor information that is representative of visitor behaviour in the reserve.

### 4.1.2 Geotag proximity to Trail Buffers

The Flickr dataset offered a larger spatial distribution of data for analysis. The trail buffer analysis showed that more geotagging occurs out with 50m of the trail network, rather than within and on the trails, as more than half of all geotags are out with the trail network buffers. A higher proportion of geotags occur within the buffers in the snow cover season, suggesting that winter trail users depend more on trail infrastructure in the reserve.

Although the results cannot be compared to existing data to confirm their accuracy, they do show that the resolution of the Flickr geotag data is high enough that it can be used to analyse proximity to trails.

### 4.1.3 Human Impact

The human impact analysis produced varied results when examining the relationship between the distance from the cluster center points and the mean ExG values. At some individual sites there was a pattern that occurred across multiple sampling years, however there was no consistent pattern of increasing or decreasing mean ExG values that was determined across all the buffer zones.

The frequency analysis revealed that between the inner 3 buffer zones (0-25m, 25-50m and 50-75m) there was only an increase in the mean ExG values between 22-40% of the time. In contrast, between the outer two buffer zones of 50-75m and 75-100m there was an increase in the mean ExG values 80 and 88% of the time, respective to the different parameters. This suggests that the greenness is more likely to increase between the buffers located furthest from the cluster center points. However, examining the difference between the innermost and outermost buffers, there is only an increase 53% of the time, representing only a very slight tendency to a higher mean ExG across the whole sampling area.

The identified cluster sites were often located near existing infrastructure such as cabins (S1, S2, S5) or viewpoints (S3), where vegetation patterns are influenced by a range of factors. Additionally, natural variation in vegetation type, topography, and soil type were not taken into consideration, but may affect the results. These variables make it difficult to isolate the effects of visitor pressure independently.

Overall the results provide limited evidence for a relationship between concentrated geotag clusters and reduced vegetation greenness (lower mean ExG values). The high frequency of increase between the outmost buffers suggests that vegetation condition may increase at the greatest distance, but there is a lack of an overall trend across all concentric buffer zones.

## 4.2 Discussion of Method

The method was chosen as it allowed for geotagged data to be compared to a zoning map that was representative of the visitor distribution patterns that had already been observed by Naturvårdsverket and Länsstryelsen. By overlaying the geotags and zoning maps, assessment of potential correlation between high visitation and areas identified as zones of concentrated use by nature reserve management could be recognised. The approach also allowed for the identification of areas where the opposite was the case, areas where visitors were frequently travelling in areas considered undisturbed in the zoning system.

Vålådalen Nature Reserve was chosen as an example due its popularity and the number of different recreational activities taking place there year round, along with the fact it has had multiple cases of land-use conflict in recent

years due to over visitation of popular sites, and between both geotag sources there was 5475 data items. The method may be less applicable in smaller nature reserves, or those in more remote locations, where smaller visitor number mean significantly smaller numbers of geotagged data.

The method used a manual collection technique for collecting the geotag data. For data collecting in one nature reserve this worked, however for geotag analysis to be an effective and efficient technique and be used across multiple nature reserves as a method of visitor pattern analysis, this method would be too slow and labour intensive.

Many studies that take place using crowd sourced geodata use API's to download data directly into the analysis interface (Wilkins et al. 2020), and adopting this methodology would increase the efficiency of data collection while reducing the time and labour requirements associated with manual extraction, which may make the method more applicable for adoption by management facilitators in protected areas.

## 4.3 Limitations

The scope of this project was limited in several ways. Due to the nature of how the geotags are recorded by the site or app user, there are limitations in the reliability of the data.

In the case of the Flickr geotag data, the technology (mobile telephone or camera) used to capture the photograph has taken the coordinates automatically based on GPS locations, but in others the user has chosen the location the photo was taken manually during the process of uploading the media onto the website. In both cases there are limitations as to the accuracy of the geotags being provided. In the case of the user entering the geotag manually, the accuracy of the geotag location relies on the memory and understanding of the user, and so accuracy will be varied. In the case of technologically automated geotagging, phone GPS is often only accurate to 5-10m, depending on the model of phone and location.

It should also be noted that the platform of Flickr is reducing in popularity (Leppämäki et al., 2025), with its peak in activity in 2011 and decreasing user rates since that time. This has implications for future studies as the Flickr

data that is available becomes less relevant, and the new data being uploaded is smaller in quantity.

Limitations exist in the Instagram geotag data, where the geotagged locations that the photos are assigned to, already exist in the application, creating "hotspots". Visitors to large area of a park or reserve therefore have all of their geotagged media assigned to exactly the same pre determined location. In Instagram, visitors always geotag photographs themselves, so again the accuracy of the data relies on the accuracy and aptitude of the user. All geotag photographs being assigned to the same geographical points limits the use of the Instagram Geotag dataset.

The adoption of techniques using geotags to monitor visitor behaviour in Swedish nature reserves could be limited to larger nature reserves, as smaller nature reserves that are less popular may have less geotag activity. Future studies could include a range of nature reserves across the country. By using social media, data limitations will also exist where the demographic that is uploading and geotagging photographs represents certain age-groups and cultures, and leans towards people partaking in specific activities, unequally across the reserve.

Examination of the trail network GIS layer revealed that the layer that SLU provides does not exactly match with the trails underneath, especially in the winter trails dataset, potentially effecting the results of the trail buffer analysis.

This study was limited by the lack of on-site visitor monitoring data that was available to correlate the visitor behaviour identified in the geotag datasets to visitor behaviour identified using other methods. It was also limited by the period of time available for data collection taking place during the snow cover season, and therefore collecting on site vegetation data for use in the human impact analysis was not possible for technical reasons.

# 5 Conclusion

This study examined if crowd-sourced geotagged data can be used to determine the behaviour of visitors in Swedish Nature Reserves, and if there is a relationship between the intensity of geotag activity and human impact within these protected areas.

The results demonstrate that geotagged data from Flickr and Instagram can provide information on visitor behaviour and distribution in protected areas. The patterns found in the density analysis of both datasets aligned with the visitor distribution expected in the documentation for Vålådalen Nature Reserve, suggesting that the data could provide a source of information for monitoring visitor behaviour and distribution.

The analysis of visitor proximity to the trail network using the Flickr Geotag dataset also determined that the data can be used to differentiate between visitors at different times of year and their proximity to the trail network at the location of their geotagged photographs. The results show that the data is extensive enough and of such a scale, that it can analysed to provide these details. Further work to statistically correlate these results to visitor monitoring statistics collected using alternate methods could improve the validity of this result.

The human impact analysis had less conclusive results, while increases in the mean ExG values and therefore the vegetation greenness could be observed between the outermost buffers, overall the pattern was inconsistent between different sample sites and years, with large Standard Deviation from the mean on multiple sampling occasions.

Overall the findings suggest that crowd-sourced geotag data can be analysed to identify visitor distribution and behaviour patterns in Swedish Nature Reserves. However the data and analysis in this case were insufficient for reliably identifying and quantifying ecological impact in relation to geotag intensity and no consistent relationship was identified.

Combined with traditional monitoring methods and with further validation through on site sampling, there is potential for geotag data to be useful as a form of data in monitoring visitors in remote protected areas.

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# Appendix A

| Location Name in Instagram Map Service | Number of Geotagged Media 2015-2025 |
|---|---|
| Lunndörrsfjällen | 318 |
| STF Lunndörrsstugan | 153 |
| Dörrsjön | 30 |
| Östra Vattensjön | 9 |
| Hottögsfjället | 26 |
| Ottfjället 1265 m. ö. h. | 816 |
| Ottfjället | 242 |
| Blanktjärn | 1193 |
| Nulltjärnarna | 1185 |
| Vålådalens Naturreservat | 258 |
| STF Stensdalsstugan | 159 |
| Vålåsjön | 6 |
| Östra Bunnertjärnen | 6 |
| Bunnerfjellene | 30 |
| Nationaal Park Vålådalen-Sylarna | 18 |
| Härjångsfjällen | 51 |
| Gåsen | 111 |
| Västra Bunnertjärnen | 6 |
| Norra Bunnersjön | 12 |
| Issjödalen | 313 |
| Pyramiderna | 122 |
| **Total** | **5064** |

# Appendix B

| Site Number | Parameter Met | Zone |
|---|---|---|
| S1 | 5 or more geotags within 50m | Active |
| S2 | 5 or more geotags within 50m | Active |
| S3 | 5 or more geotags within 50m | High Active |
| S4 | 5 or more geotags within 100m | Restricted |
| S5 | 5 or more geotags within 100m | Active |

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